Presenter notes

Slide 1

Timing: 0:00–2:30. The leadership story starts with an employee trying to find the right rule for a particular situation.

Branch employees often need a specific answer from a long procedure while they are helping someone. Finding a keyword is only part of the task. The employee needs the applicable action, the conditions that make it applicable, and any exception that changes the outcome. The business objective is faster access to correct guidance, with evidence the employee can inspect. No time savings or productivity uplift have been measured in this exercise.

I built a functional conversational assistant over all eleven supplied Cayman Bank procedures. The collection includes 139 HTML tables and produces 1,388 traceable passages. The example on the right is a real source fact and a recorded tool outcome: the overnight limit is $5,250 in a single coin-counting machine. The employee can open the original passage rather than taking the fluent answer on trust.

Three ideas hold the solution together. First, keep the rule attached to its situation during document preparation. Second, retrieve relevant source material before asking a language model to write an answer. Third, make sources, checks and appropriate fallback part of the employee experience. A follow-up starts a fresh evidence search; a previous generated answer is never treated as proof.

The delivered prototype includes actual local language-model inference, a conversational workspace, source inspection, procedure search, feedback, exports and measured evaluation. It does not execute transactions or establish that the supplied snapshot is current production policy. The Wells Fargo-inspired visual treatment is explicitly an interview prototype over the supplied Cayman Bank documents.

Generative AI assisted requirements analysis, implementation, tests and presentation preparation, and powers runtime answers. The candidate owns the review and must be able to explain the decisions personally. AI-authored evaluation cases are useful regression material, not independent ground truth. The recommendation is a scoped validation phase with procedure owners before any operational pilot.

Transition: to understand why this answer can be checked, first look at how the original rules were organized.

Slide 2

Timing: 2:30–6:00. Continue the coin-machine example to explain the data choices, not simply the tools.

A monthly task is not enough context to choose a control level. In PR63, monthly testing is single control, while weekly balancing and random monthly cash counts use dual control. If extraction separates the action from its row label, the generator can retrieve the right words and still give the wrong instruction. This is why the content design comes before model selection.

The eleven input files are HTML exports with meaningful policy classes. The parser recognizes titles, publication text, sections and subsections, and explicitly decodes UTF-8. It removes scripts and presentation-only material, never executes source HTML, and does not fetch linked procedures. Linked material is outside the supplied evidence boundary.

Tables are serialized with column labels, row conditions and positions. Rowspans carry their conditions forward; colspans preserve alignment; nested table content is included once. Oversized row continuations repeat the leftmost condition. This reduces detachment of a requirement from its situation, but does not prove that every complex table has been semantically interpreted correctly.

Narrative passages target roughly 110 whitespace words within a section. Long blocks overlap by 20 words. Unsplit table rows can reach 130 words, and a repeated condition can make a continuation longer. These are word targets. The generator separately enforces its prompt-token budget. An earlier larger grouping diluted specific instructions, so shorter passages were selected after observing development failures.

Every passage retains its procedure ID, source filename and hash, section, publication text, anchor and table location. The original eleven files remain unchanged. Identical input produces stable passage identities; changed input requires rebuilding the index. Extraction diagnostics found all checked paragraph five-word sequences, which is evidence of text preservation, not proof of policy interpretation.

Search then combines complementary strengths: BM25 for exact identifiers and specialized wording; MiniLM vectors for paraphrases; reciprocal rank fusion to combine positions; and a cross-encoder to judge question and passage jointly. It selects up to three nonduplicate passages, targeting about 520 words. Ranking scores choose evidence; they are not probabilities that an answer is correct.

Transition: the next slide follows that evidence through the live request, including the release and fallback decisions.

Slide 3

Timing: 6:00–9:30. Explain the request in order, then defend the choices and their limits.

A request first passes basic length, identifier and direct-instruction checks. The resolver identifies an explicit procedure, a new named topic or a continuation of the user's situation. Search scope is enforced before retrieval. Sources and cached index artifacts are checked for changes, including the vector-matrix hash, dimensions and normalization.

BM25 and normalized MiniLM embeddings search the same 1,388 passages. Reciprocal rank fusion combines their top candidates without comparing incompatible raw scores. The ms-marco MiniLM cross-encoder reranks the fused candidates. Weak retrieval signals lead to clarification or insufficient support. The small collection fits exact in-memory vector search, so a managed vector database would add infrastructure without demonstrated benefit here.

Relevant source passages, including a bounded adjacent qualification when applicable, become the evidence pack. Qwen3-4B in Q4_K_M GGUF form generates locally through llama.cpp on CPU. Non-thinking operation, greedy decoding, a 3,800-input-token limit and a 240-output-token limit suit this constrained answer task. The prompt asks for the exact fact first, relevant conditions and citations, using only evidence. Greedy decoding does not guarantee correctness.

Qwen3 was selected after development comparisons with a smaller Qwen2.5 baseline that produced more distracting detail and withheld drafts. This was an end-to-end engineering iteration, not an isolated causal experiment. Local inference avoids API-key and hosted inference dependencies for the exercise, while adding CPU latency. A stronger approved deployment model still needs comparison before a pilot.

Checks bind claims to their actual citations, compare numeric values and recognizable thresholds, and screen support. Substantive exact wording is matched to the cited passage; paraphrases use a DeBERTa NLI model. Its thresholds are development settings, not calibrated confidence. A failed draft gets at most one correction with the same evidence and failed-check feedback. If it still fails, the draft is withheld and source text remains available. The screen can also reject a correct answer or miss an error.

This is one observable orchestrator with read-only evidence access, not an autonomous bank transaction agent. Activity records show real actions, counts and timings, not private model reasoning. RAG was chosen over fine-tuning for source replacement and traceability; sending every document would waste context and mix unrelated conditions.

Transition: these controls matter only if an employee can understand and use them in conversation.

Slide 4

Timing: 9:30–13:00. Read the short conversation and point to the selected evidence, then explain the broader experience.

The dialogue reproduces actual recorded application responses. First the employee asks about the overnight coin-machine limit. The assistant answers $5,250 and cites S1. The employee then asks whether monthly testing is single or dual control. The assistant retrieves the monthly-testing instruction and answers single control with S2. The source crop on the right is an unaltered browser capture from the approved recording, displayed through a native PowerPoint crop.

The useful behavior is fresh evidence for a different task within the same procedure. The resolver retains PR63 as the topic but does not dilute testing retrieval with overnight-storage wording. Citation buttons select the exact passage for that particular answer. The employee can read the section, date and surrounding conditions or download the original procedure.

Different follow-ups require different handling. After an emailed-document notary question, saying that the requester is physically present must retain the emailed-document situation. The supplied rule still prohibits it. A hypothetical repeating the notary topic or PR1000 now preserves that original condition too. An explicitly different topic, such as savings-bond fees, replaces old scope and retrieves a new procedure. This is bounded deterministic context handling; ambiguous long conversations remain a limitation.

The workspace includes guided starters, optional procedure scope, a searchable library and an Evidence desk. Evidence shows original text and source identity. Checks exposes citation, amount, threshold and support screening. Activity shows the actions that actually ran and their timings. A failed draft can be corrected once; a still-failing draft is withheld while original evidence stays available. The interface does not show a fabricated certainty score.

Search scope is a real restriction: a question about a different selected procedure prompts clarification. New investigation clears conversation context. Browser sessions keep histories isolated, while model objects are cached for reuse. The employee can mark a response for review and export the conversation with sources, checks, activity and feedback. These are session features, not an implemented enterprise audit service.

The demo later shows these cases in the live tool or the approved silent recording. Source-code editors and terminals are excluded, as required by the assignment.

Transition: now distinguish a convincing interaction from evidence that the solution answers reliably.

Slide 5

Timing: 13:00–17:00. This is the evidence slide. Separate passage retrieval, answer diagnostics, coverage and independent correctness.

The suite contains 30 ordinary questions covering all eleven procedures, four two-turn conversations and six selected boundary cases. Nineteen ordinary questions are development cases and eleven are challenge cases. Labels were AI-authored from the supplied documents and source-checked; after development inspection, the challenge split is not a blind external holdout. Results describe this regression set, not enterprise accuracy.

For the same questions, keyword search finds a labelled relevant passage in the first five results for 27 of 30 cases, semantic search for 28, and hybrid retrieval plus reranking for all 30. Hybrid mean reciprocal rank at five is 0.9278. Finding one labelled passage does not establish that every material exception was retrieved or that the generated answer is correct.

The automatic expected-fact proxy passes 23 of 30 ordinary cases. It checks phrase groups, so it can miss a correct paraphrase and pass a wrong statement containing expected terms. A source-only fallback counts as a failure of generated-answer coverage even if the excerpts contain the answer. Across 35 generated evaluations, six drafts were withheld after checks. The four original final-turn follow-up proxies include two

misses; those misses remain in the report. Six deliberately chosen boundary cases matched their expected statuses, which is not a general security assessment.

The table makes the four requested quality dimensions explicit. Accuracy needs an independent reviewer to assess every instruction against the source and stated situation; citations, numeric consistency and NLI are only diagnostics. Relevance asks whether the exact question is answered without unrelated workflow detail. Completeness asks whether applicable conditions, exceptions, warnings and escalation requirements are present. Conciseness asks whether wording is brief without becoming incomplete. The measured mean draft length is 28 words across 32 drafts with recorded length, including withheld drafts; it is not a measure of answer usefulness.

Serial local CPU latency is p50 13.81 and p95 23.92 seconds, excluding initial model load. The engineering suite has 49 passing tests for mechanics and safeguards, and real browser workflows were checked. Neither test count nor latency proves substantive correctness.

Next, two operational subject-matter reviewers should independently score all four dimensions on unseen scenarios, adjudicate disagreements, and report critical errors separately from average scores. Include false refusals, answer coverage, source-version conflicts and long conversations. Compare employee task success and time to correct guidance against existing search before claiming business value.

Transition: these findings shape both what we will demonstrate now and what leadership should validate next.

Slide 6

Timing: 17:00–20:00, followed by the separate 15-minute technical walkthrough. Close the leadership story with a concrete demonstration and decision.

We started with an employee trying to find the applicable branch rule. We preserved the rule's situation during ingestion, retrieved evidence before generation, exposed the checks and tested retrieval and response quality separately. The working prototype demonstrates the approach; independent validation determines where it is safe and useful.

The walkthrough will use the actual application or the clearly labelled approved silent recording. First, ask the overnight coin limit and follow up about monthly testing. Inspect both source passages to show that a different task uses fresh evidence. Next, ask about notarizing an emailed document and change the condition by making the requester physically present. The prohibition remains. Finally, ask the savings-bond limit for an account younger than twelve months, then the Private Bank Experience exception. The source's $1,000 daily aggregate limit has a stated exception; in the approved recording, the generated exception draft fails a check after one correction, so the tool retains the original source text for review. Do not present that fallback as a successful generated answer.

During the walkthrough, also show a topic switch, missing context after reset, unsupported requests, feedback and the evidence export. Share the application and slides, never implementation source. The approved 9:25 recording fits within the 15-minute allocation and leaves time for live source inspection and questions. Keep the local model warm and the recorded backup available; a recording must be identified as such.

The recommendation is a scoped validation phase. Procedure owners and two independent subject-matter reviewers should evaluate unseen scenarios across accuracy, relevance, completeness and conciseness. Critical unsupported instructions need explicit investigation and a release gate; coverage and unnecessary refusal must stay visible. The business and validation teams must agree acceptable thresholds rather than treating the development metrics as achieved enterprise service levels.

Branch operations and AI engineering should measure task success and time to correct guidance against current search, and compare stronger approved models with realistic latency and cost. Before an operational pilot, security and platform owners need employee identity, document entitlements, authoritative source versions, approved audit and retention, monitoring, escalation and rollback. These are future production requirements, not completed features of the exercise.

Generative AI assisted design, implementation, evaluation drafting and these materials; local Qwen generation powers the tool. Candidate understanding, accurate disclosure and ownership remain essential. The six-slide presentation and its supporting PDFs exclude implementation source, and the approved final video remains preserved unchanged.

Handoff: move into the prepared walkthrough and ask the first coin-machine question.